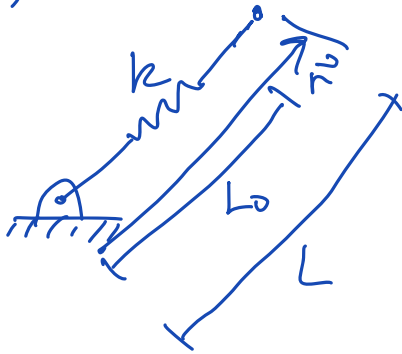


News

# Particle Dynamics

- Review Spring
- EOM - Zero Length Spring Ex.
- Numerical Approach to  $\vec{r}, \vec{v}, \vec{a}, t$

Ass. Survey today 3pm  
HW1 Friday 6pm.

Review Spring (2D, no gravity) $\hat{i}, \hat{j}$ FBD

$$\begin{aligned}\vec{F} &= (L - L_0)k \left( \frac{-\vec{r}}{|\vec{r}|} \right) \\ &= (|\vec{r}| - L_0)k \left( \frac{-\vec{r}}{|\vec{r}|} \right)\end{aligned}$$

This is an example of a coupled eq<sup>n</sup>.

Solve LMB  $\Rightarrow \sum F_i = \dot{\vec{p}}$

Rearrange  $\Rightarrow m\vec{a} = \vec{F}$

$$m(\ddot{x}\hat{i} + \ddot{y}\hat{j}) = -k \left[ \frac{(\sqrt{x^2 + y^2} - L_0)(x\hat{i} + y\hat{j})}{\sqrt{x^2 + y^2}} \right] \quad (1)$$

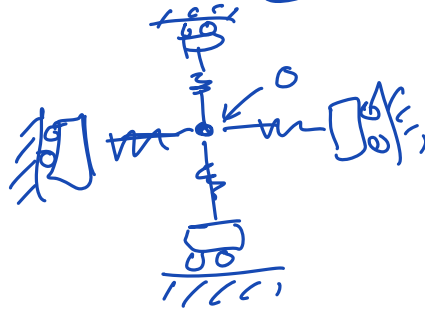
For x-direction, eval (1)  $\cdot \hat{i}$

$$m\ddot{x} = -k \left[ \frac{(\sqrt{x^2 + y^2} - L_0)x}{\sqrt{x^2 + y^2}} \right]$$

y appears in eq<sup>n</sup> for x  $\therefore$  coupled  $\rightarrow$

# Special Case: The zero length spring.

$$L_0 = 0$$



Eval,  $\hat{i}$  &  $\hat{j}$  directions

$$\left. \begin{aligned} m\ddot{x} &= -kx \\ m\ddot{y} &= -ky \end{aligned} \right\} \text{decoupled system.}$$

Convert to first order form  $\Rightarrow$  Equations of Motion

$$\left. \begin{aligned} \dot{x} &= v_x & \dot{v}_x &= -\frac{k}{m}x \\ \dot{y} &= v_y & \dot{v}_y &= -\frac{k}{m}y \end{aligned} \right\} \textcircled{2}$$

Take this one step further.

$$\begin{aligned} X &= \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix} \Rightarrow \begin{aligned} \dot{x} &= v_x \\ \dot{y} &= v_y \\ \dot{v}_x &= -\frac{k}{m}x \\ \dot{v}_y &= -\frac{k}{m}y \end{aligned} \end{aligned}$$

$\uparrow$   
state

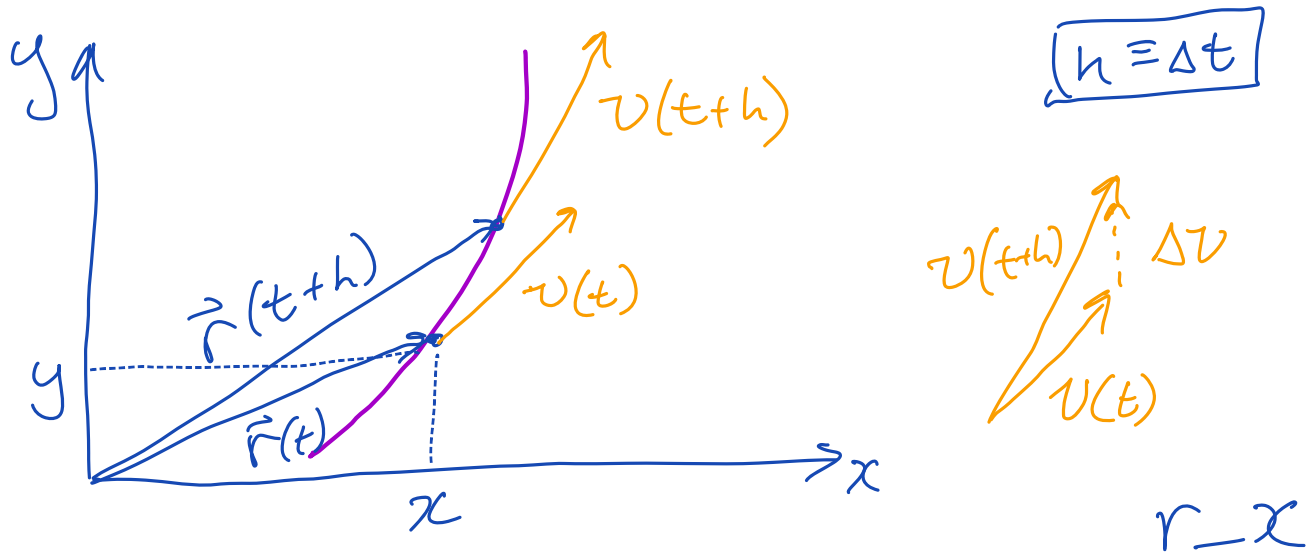
system 1st order  
linear odes.

$$\Rightarrow \dot{X} = f(X)$$

$\uparrow$   
rate of change  
as function of  
state.

$$\boxed{X_i} \rightarrow f(x) \rightarrow \boxed{\dot{X}_i} \rightarrow ? \rightarrow \boxed{X_{i+1}}$$

Numerical Approach to  $\vec{r}, \vec{v}, \vec{a}, t$  TLDR.



$$\vec{r} = x\hat{i} + y\hat{j} = r_x\hat{i} + r_y\hat{j}$$

We can approximate the velocity as

$$\vec{v}(t) = \frac{d\vec{r}}{dt} \simeq \frac{\Delta\vec{r}}{\Delta t} = \frac{\vec{r}(t+h) - \vec{r}(t)}{h} \quad (3)$$

Similarly

$$\vec{a}(t) = \frac{d\vec{v}}{dt} \simeq \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}(t+h) - \vec{v}(t)}{h} \quad (4)$$

Rearrange (3) & (4)

$$\begin{aligned}\vec{r}(t+h) &= \vec{r}(t) + h\vec{v}(t) \\ \vec{v}(t+h) &= \vec{v}(t) + h\vec{a}(t)\end{aligned}\quad \left. \vphantom{\begin{aligned}\vec{r}(t+h) &= \vec{r}(t) + h\vec{v}(t) \\ \vec{v}(t+h) &= \vec{v}(t) + h\vec{a}(t)\end{aligned}} \right\} \text{Euler's Method.}$$

future  
state  
 $x_{i+1}$

current  
state  
 $x_i$

current  
ROC.  
 $h\dot{x}_i$

if we know current state, we can  
predict the future







